Ansh Pethani

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EDUCATION

North Carolina State University, Raleigh, NC, USA — **Aug 2025 - May 2027**
Master of Computer Science — *GPA: 4/4*

Vellore Institute of Technology, Vellore, IN — **Sep 2021 - May 2025**
Bachelor of Technology in Information Technology — *GPA: 8.91/10*

EXPERIENCE

Research Associate, *iEXCEL Lab, CS Department, NC State University, USA* — **Aug 2026 - Present**

- Managing CI/CD pipelines, backend deployments, API integrations, and web features.
- Designed a modular BERTopic pipeline benchmarking 5 embedding models on 3,200+ publications, combining coherence/diversity metrics with LLM evaluation to drive model selection for a multi-year research collaboration study.
- Engineering a GNN to validate trophic interactions and generate interaction plausibility scores for food web data.
- Integrating pre-trained multi-species computer vision pipelines into existing backend and frontend architectures.
- Review and approve work from undergraduate researchers and give feedback to improve it.

Research Associate, *IEC Lab, CS Department, NC State University, USA* — **Aug 2025 - Aug 2026**

- Developed Python ETL pipelines to parse, clean, and resolve various conflicts across CSV, XML, & JSON records.
- Developed and optimized an LLM-based Python pipeline for automated quiz generation using student responses.
- Researched embedding models and used NV-Embed-v2, evaluating them using K-Means clustering.

AI Development Intern, *IoTIoT***,** India — **Jan 2025 - Jul 2025**

- Developed AI/ML systems, including an offline chatbot running locally (Ollama, Flask, Tauri, Rust), a fine-tuned Hugging Face model for zero-shot voice cloning, and image detection models using TensorFlow/Keras
- Built a Google A2A (Agent-to-Agent) wrapper integrating WhatsApp with MCP (Model Context Protocol) and engineered generative AI applications using Gemini/OpenAI APIs, documenting technical learnings through AI-focused blogs.

Python Backend and AI Development Intern, *DeepKlarity***,** India — **Sep 2024 - May 2025**

- Engineered ML & AI solutions, including YOLOv11-based image detection models (dog-nose & cat-face) and generative AI-powered chatbots (blog generation, RAGs for document/PDF/image Q&A).
- Built and deployed applications across domains: Gmail add-on for automated email risk assessment, AutoEDA tool for PDF-based data analysis, and a data analytics platform with an admin dashboard using Python and generative AI.
- Designed and built full-stack web applications using React, JavaScript, FastAPI, and PostgreSQL, developing REST APIs and AI-powered features that improved automation, scalability, and client usability.

PROJECTS

- **Hungry Wolf - Gamified Food Delivery Platform (2025):** Built the delivery module end-to-end (backend routes/models and frontend order-management UI), tested and integrated the final build, and fixed the CI/CD pipeline for a full-stack, gamified food-ordering platform. *Tech Used:* TypeScript, ReactJS, Node.js, Express.js, REST APIs, Firebase
- **IDS Attack Detection in IoV CAN Bus (2024):** Benchmarked and optimized 12 ML/DL models for accuracy and runtime efficiency on a 1.4M-instance dataset, applying SMOTE for class imbalance. *Tech Used:* Python, TensorFlow, Scikit-learn, Imbalance-learn, Pandas, NumPy
- **Real-Time Analysis (2024):** Developed a data processing app to analyze web-scraped data at 5-second intervals, featuring automatic graph visualizations as new data batches were received; *Tech Used:* Python, PySpark, Hadoop
- **NBA/ABA Statistics and Future Predictions (2025):** Performed data mining, association mining, and applied ML models to analyze and predict game and player performance outcomes. *Tech Used:* Python, Scikit-learn, Pandas, NumPy

SKILLS & TOOLS

- **Languages:** Python, SQL, TypeScript, JavaScript, Java, C++, C
- **AI/ML:** OpenAI API, Gemini API, Ollama, Hugging Face, PyTorch, TensorFlow, Scikit-Learn, Keras, NumPy, Pandas
- **Frontend:** ReactJS, HTML, CSS | **Backend:** FastAPI, Flask, REST APIs | **Databases:** Postgres, Firebase Firestore, ChromaDB
- **Tools:** GenAI, Git, Linux, Postman, Excel, VS Code, Cursor, Claude Code, Docker, Rust, Tauri, MATLAB, ROS2, Gazebo
- **Coursework:** AI, ML, SWE, DBMS, DSA, Design and Analysis Of Algorithms, Automated Learning & Data Analysis, Cloud Computing, CN, Computer & Network Security, Information Security, Cyber Security, Network Information Security, Advanced Robotics, Computer Architecture, OS, Parallel Systems

PUBLICATIONS

- Pethani, A., Nallakaruppan, M.K. (2024). "Electronic Copyright and Legal Application Regarding Non-Fungible Tokens (NFTs)." Springer, Advances in Intelligent Systems and Computing, vol. 1457. [Link]
- Nallakaruppan, M.K., Pethani, A., Pelusi, D. (2025). "Data Security in the Age of Marketing: Safeguarding Customer Information and Compliance." Emerald Publishing, Data Engineering for Data-driven Marketing, pp. 165–178. [Link]